Table 4: Model fitting summary of the logistic regression models used for matching in each Cursor adoption cohort.

Cohort	Candid. Repos	AUC	McFadden's Pseudo R^2
202408	807,608	0.8506	0.1412
202409	804,740	0.9137	0.2336
202410	818,160	0.8969	0.2204
202411	796,312	0.9144	0.2722
202412	794,186	0.8907	0.2101
202501	776,547	0.8702	0.2123
202502	762,968	0.8716	0.2271
202503	792,382	0.8281	0.1939

A Matching Results

In this section, we present additional details about the propensity score matching process described in Section 3.1.3.

Table 4 presents the total number of candidate repositories in each major Cursor adoption cohort and the AUCs and McFadden's Pseudo R^2s from fitting each logistic regression model in that cohort with 10,000 sampled candidates. In general, the logistic regression models achieve very high goodness-of-fit (0.8281 to 0.9144 AUC), but only explain a relatively limited amount of variance in the outcome (14.12% to 27.22%). This indicates that while the models can distinguish Cursor adoptions relatively well, the measured covariates alone do not fully explain the variations behind each adoption case. We view this as an expected result, as the Cursor adoption decisions are probably not driven by any observable information in GitHub; the latter may only latently and partially capture it to help us create a reasonably comparable control group.

To assess whether the treatment and control group repositories are reasonably comparable, we plot the distribution of propensity scores (Figure 5) and conduct balance checks on observable covariates before each adoption (Table 5). While the propensity score plot (Figure 5) shows that the treatment and control group repositories have highly similar conditional treatment probabilities, the matched control group is slightly skewed toward lower propensity scores. Furthermore, the balance checks (Figure 5) indicate acceptable but imperfect matching: While all metrics show acceptable balance between the treatment and control group in terms of normalized differences (|Norm. Diff| < 0.25), there are still observable mean differences between the two groups. In other words, it indicates that our matching is imperfect and our study setting deviates from perfect randomized controlled experiments. This motivates us to adopt a difference-in-difference design for causal inference instead of simply comparing outcomes between the two groups—a DiD design with two-way fixed effects and time-varying covariates is more suited for a quasi-experimental setting like ours.

B Alternative DiD Estimators

Recall from Section 3.3 that several alternative DiD estimators are available for estimating the average treatment effect on treated ATT and the "horizon-average" treatment effect ATT_h. In this section, we provide a brief introduction to the two other widely popular alternative estimators, namely the two-way fixed effects (TWFE) estimator and the Callaway and Sant'Anna [31] estimator. Then, we compare their estimation results with results from the Borusyak et al. [29] estimator presented in the main paper.

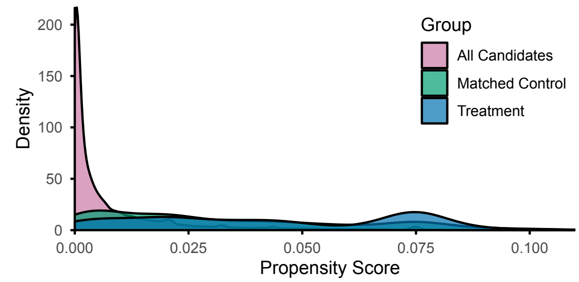

Figure 5: The distribution of propensity scores between all candidate repositories, matched control group repositories, and the 806 Cursor-adopting repositories, showing that the matched control group has highly similar conditional treatment probabilities compared to the treatment group.

Table 5: Balance statistics: treatment versus matched control pre-adoption. Normalized difference is defined as $(\bar{X}_t - \bar{X}_c)/\sqrt{(S_t + S_c)/2}$, where $\bar{X}_t/\bar{X}_c$ and S_t/S_c stands for the mean and variance in the treatment/control group, respectively. Follwing balance check conventions [24, 106], we consider |Norm. Diff| < 0.25 as acceptable balance and < 0.1 as good balance. Note that the treatment means here are different from the treatment means in Table 1 because the latter reflect the metrics *at the time of data analysis* while the Table here reflects the same metrics *at the time of Cursor adoption.*

Metrics	Treatment Mean	Control Mean	Norm. Diff
Age (in days)	496.07	681.38	−0.207
Comments	2870.82	564.84	0.147
Forks	196.07	64.27	0.079
Issues	524.07	103.95	0.154
Pull Requests	1075.58	266.03	0.158
Releases	27.51	25.44	0.006
Stars	1056.65	334.62	0.130
Total Events	10103.25	2443.39	0.171
Users Involved	1247.79	414.90	0.137

B.1 The Two-Way Fixed Effects Estimator

The TWFE estimator estimates ATT from the $\hat{\beta}$ parameter in the following ordinary least squares regression (OLS) with per-repository (μ_i) and per-period (λ_t) fixed effects (thus "two-way"), *on all available observations* with $D_{it} = 1$ for treated and $D_{it} = 0$ otherwise:

$$Y_{it} = \hat{\mu}_i + \hat{\lambda}_t + \hat{\beta} D_{it} + \hat{\Gamma}' Z_{it} + \epsilon_{it} \tag{8}$$

For ATT_h and pre-trend test parameters (Equation 5), the TWFE estimator typically estimates one single OLS regression in the following form, *on all available observations*:

$$Y_{it} = \hat{\mu}_i + \hat{\lambda}_t + \hat{\Gamma}' Z_{it} + \sum_{h=-k, h \neq -1}^{j} \hat{\tau}_h \mathbf{1}[t = E_i + h] + \epsilon_{it} \tag{9}$$